

Amplifier Terms:

Amplifiers can be described on their

- 1) class of operation
- 2) Types of coupling
- 3) Ranges of frequency.

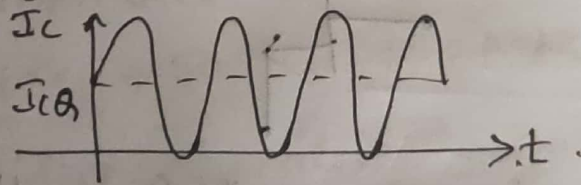
Classes of operation:

There are 3 classes of operation:

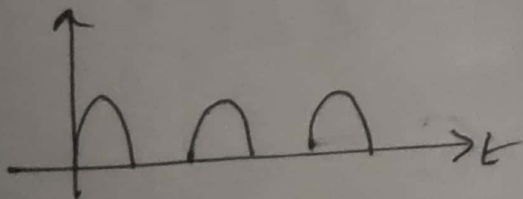
Class A operation:

→ In class A operation, an amplifier i.e., transistor operates in active region.

→ collector current flows for 360° ac cycle & dc operating point will be near the middle of the load line.

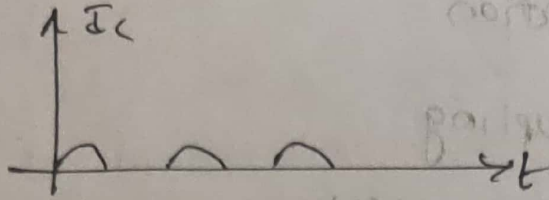
Class B operation:

→ the collector current flows only for half cycle (180°) & a point is at cutoff. Only the half of ac base voltage can produce collector current.



3) class C operation:-

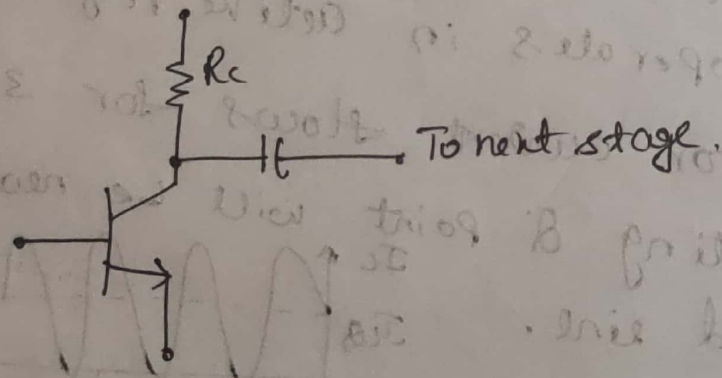
→ collector current flows for less than 180° of the ac cycle. only part of the half cycle of ac base voltage produces collector current.



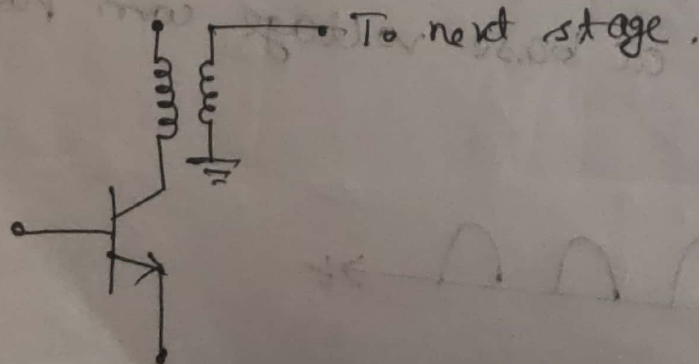
2) Types of coupling:-

There are three types of coupling

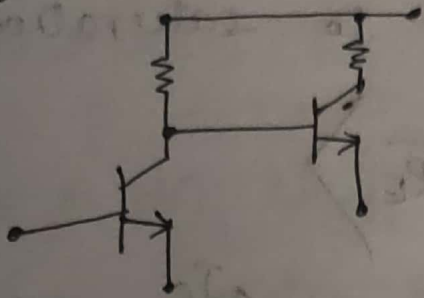
1) capacitive coupling:- The coupling capacitor transmits the amplified ac voltage to the next stage & blocks dc.



2) Transformer coupling:- Here the ac voltage is coupled through a transformer to the next stage & blocks dc.



3) Direct coupling: Here direct connection b/w the collector of the first transistor & the base of the second transistor. Hence both ac & dc voltages are coupled.



3) Range of frequency:

→ An audio Amplifier operates in the range of 20Hz to 20kHz .

→ RF amplifier amplifies frequencies above 20kHz .

i) AM radios $\Rightarrow$ b/w 535 & 1605kHz

ii) FM radios $\Rightarrow$ b/w 88 & 108MHz

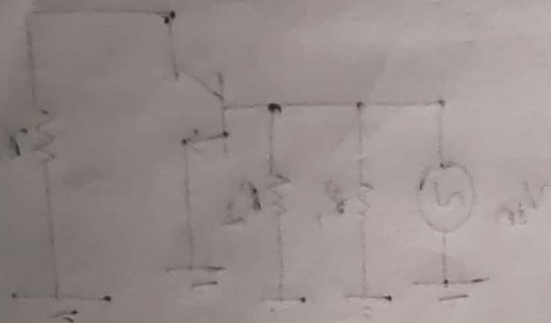
→ A narrowband amplifier works over a small freq range like 450 to 460kHz .

→ A wideband amplifier operates over 0 to 1MHz .

Two load lines:

i) DC load line

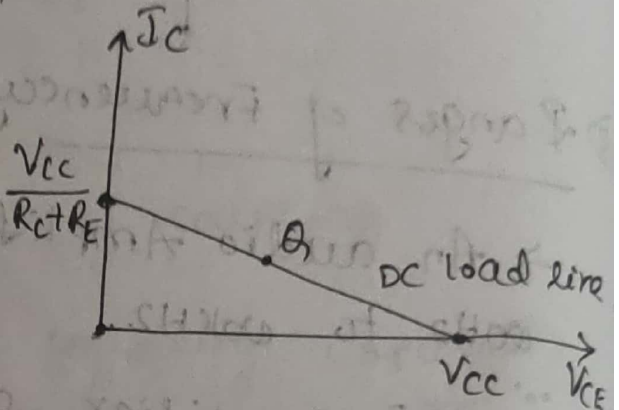
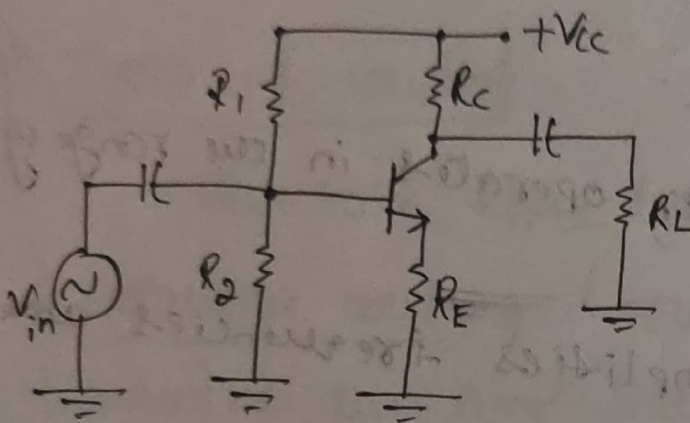
ii) AC load line



→ DC load line:

→ Fig. shows the VDB amplifier. Q point can be varied by varying R_2 . For large values of R_2 transistor goes to saturation

$$\therefore \langle I_{CCsat} = \frac{V_{CC}}{R_C + R_E} \rangle$$

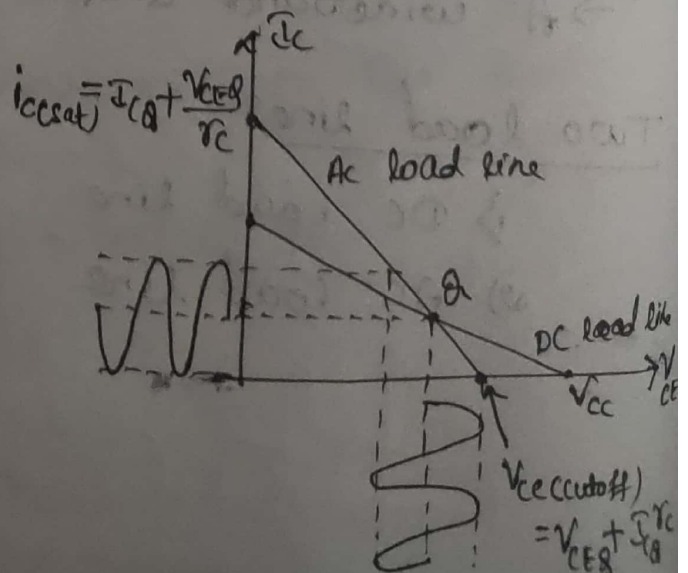
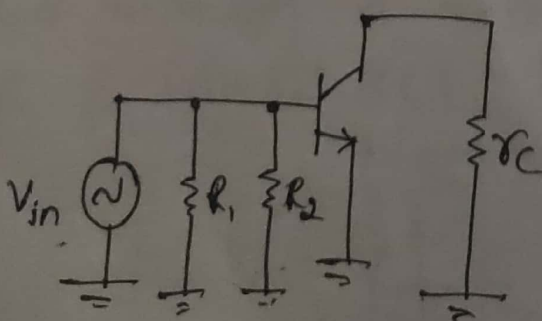


→ For small values of R_2 , transistor goes into cutoff. v_{BE} is

$$\langle V_{CE(cutoff)} = V_{CC} \rangle$$

② AC Load line

→ Fig. below shows the AC equivalent of VDB amplifier.



→ With emitter is at ground, R_E has no effect on the ac operation. &

$$(r_c < R_c)$$

→ since ac collector & emitter resistance are lower than dc resistance, the ac load line is much steeper.

→ Both ac & dc lines intersect at Q point.

→ considering collector vgl loop.

$$V_{ce} + i_c r_c = 0$$

$$\text{i.e., } i_c = -\frac{V_{ce}}{r_c} \quad (1)$$

→ ac collector current is

$$i_c = \Delta I_c = I_c - I_{cQ}$$

→ ac collector vgl is

$$v_{ce} = \Delta V_{ce} = V_{ce} - V_{ceQ}$$

$$\therefore \text{eqn (1)} \Rightarrow I_c = I_{cQ} + \frac{V_{ceQ}}{r_c} - \frac{V_{ce}}{r_c}$$

Considering transistor in saturation, $V_{ce} = 0$

$$\therefore i_{c(sat)} = I_{cQ} + \frac{V_{ceQ}}{r_c}$$

where, $i_{c(sat)}$ = ac saturation current

I_{cQ} = dc collector current

V_{ceQ} = dc collector-emitter voltage

r_c = ac collector resistance.

→ when transistor in cutoff,

$$I_C = 0.$$

$$V_{CE}(\text{cutoff}) = V_{CEQ} + \Delta V_{CE}$$

$$\text{WKT, } \Delta V_{CE} = (\Delta I_C)(r_C)$$

$$\therefore \Delta V_{CE} = (I_{CQ} r_C)$$

$$\therefore \langle V_{CE}(\text{cutoff}) = V_{CEQ} + I_{CQ} r_C \rangle$$

→ when a point is below center of ac load line,

$$\text{maximum peak, } \langle m_P = I_{CQ} r_C \text{ or } V_{CEQ} \rangle.$$

& maximum peak to peak output is,

$$\langle m_{PP} = 2m_P \rangle$$

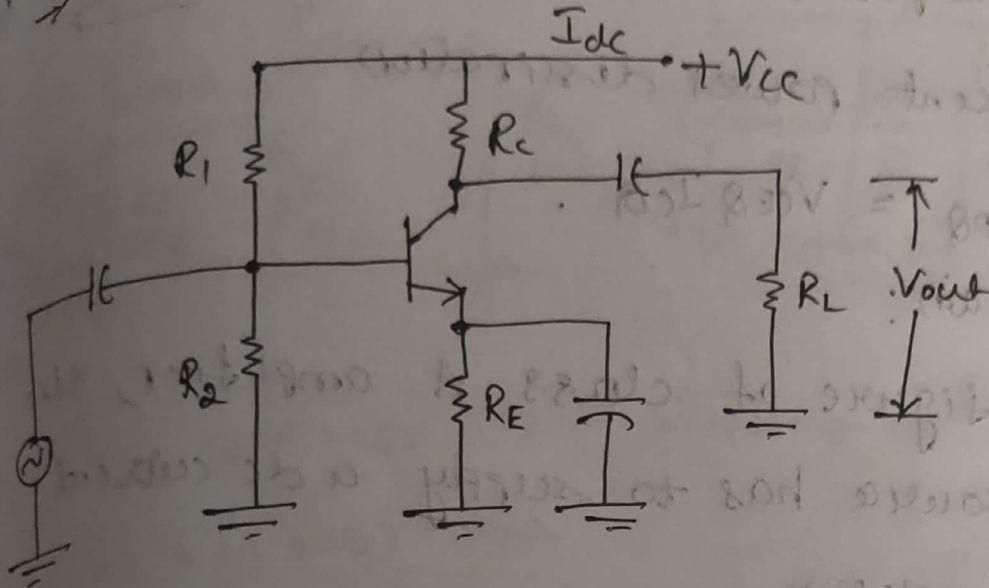
→ when Q-point is at the center of the ac load line,

$$\langle I_{CQ} r_C = V_{CEQ} \rangle$$

$$\therefore \left\langle R_E = \frac{R_C + r_C}{\frac{V_{CE}}{V_{E}} - 1} \right\rangle$$

class A operation:

- fig below shows the VCB Amplifier ie is a class A amplifier.
- collector current flows throughout the cycle.



Analysis of class A Amplifier:

1) power Gain:

$$A_p = \frac{P_{out}}{P_{in}} = \frac{\text{ac output power}}{\text{ac input power}}$$

2) output power:

in rms volts,

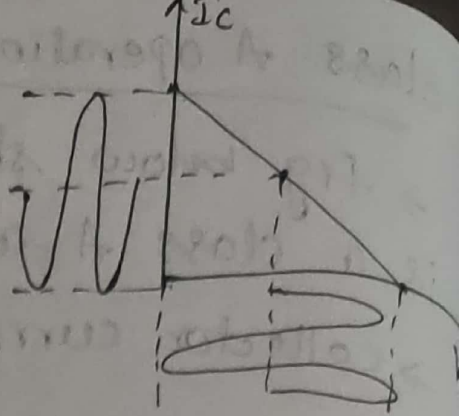
$$P_{out} = \frac{V_{rms}^2}{R_L}$$

in peak-to-peak volts

$$P_{out} = \frac{V_{out}^2}{8R_L}$$

3) maximum output power:

$$P_{outmax} = \frac{m_{pp}^2}{8R_L}$$



4) Transistor power dissipation.

Quiescent power dissipation

$$P_{DQ} = V_{CEQ} I_{CQ}$$

5) Current Drain:

→ From figure of class A amplifier, the dc voltage source has to supply a dc current I_{DC} to the amplifier.

→ This dc current has two components

i) The biasing current through the voltage divider

ii) the collector current through the transistor.

→ ∴ the dc current is called current drain of the stage.

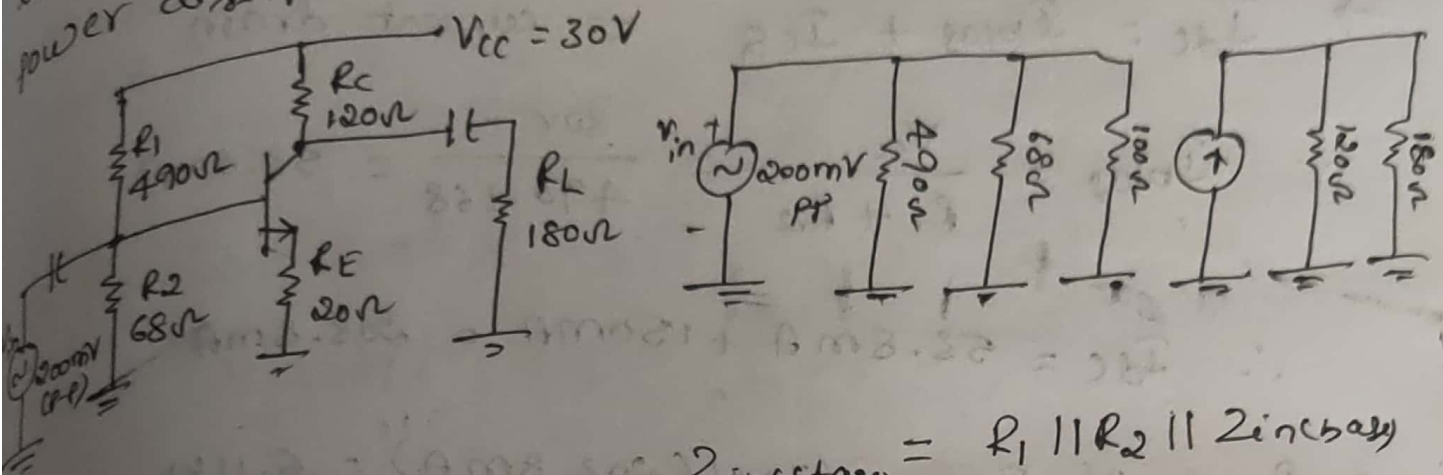
6) Efficiency:

The dc power supplied to the amplifier by the dc source is

$$P_{DC} = V_{CC} I_{DC}$$

$$\therefore \text{efficiency, } \left(\eta = \frac{P_{out}}{P_{DC}} \times 100\% \right)$$

Ex): For the circuit shown, if the peak-to-peak output voltage is 18V & the input impedance of the base is 100Ω, calculate power gain, transistor power dissipation & efficiency?



soln: $MPP = 18V$

$Z_{in(base)} = 100\Omega$

$$Z_{in(stage)} = R_1 \parallel R_2 \parallel Z_{in(base)}$$

$$= (490) \parallel (68) \parallel (100)$$

$$= 37.4\Omega$$

$\rightarrow P_{in} = \frac{V_{in}^2}{8 R_L} \quad (R_L = Z_{in(stage)})$ at i/p stage

$$= \frac{(200mV)^2}{8 \times 37.4} = 133.7 \mu W$$

$\rightarrow P_{out} = \frac{MPP^2}{8 R_L} = \frac{(18V)^2}{8 (180\Omega)} = 225 mW$

$\rightarrow$ power gain, $A_p = \frac{P_{out}}{P_{in}} = \frac{225 mW}{133.7 \mu W} = 1683$

$\rightarrow I_E = \frac{V_E}{R_E} = \frac{3V}{20\Omega} = 150mA$

$V_{BB} = \frac{R_2}{R_1 + R_2} V_{CC} = \frac{68}{68 + 490} (30)$

$V_{BB} = 3.65V$

$\rightarrow V_C = V_{CC} - I_C R_C$

$= 30 - (150m)(120\Omega)$

$= 12V$

$V_E = V_{BB} - V_{BE} = 3.65 - 0.7$

$V_E = 3V$

$$\rightarrow V_{CEQ} = V_C - V_E = 12V - 3V = 9V$$

$$\rightarrow P_{DQ} = V_{CEQ} I_{CQ} = (9V)(150mA) = 1.35W$$

$$\rightarrow I_{DC} = I_{bias} + I_{CQ} \quad \leftarrow \text{current drain}$$

$$\therefore I_{bias} = \frac{V_{CC}}{R_1 + R_2} = \frac{30V}{490 + 68} = 53.8mA$$

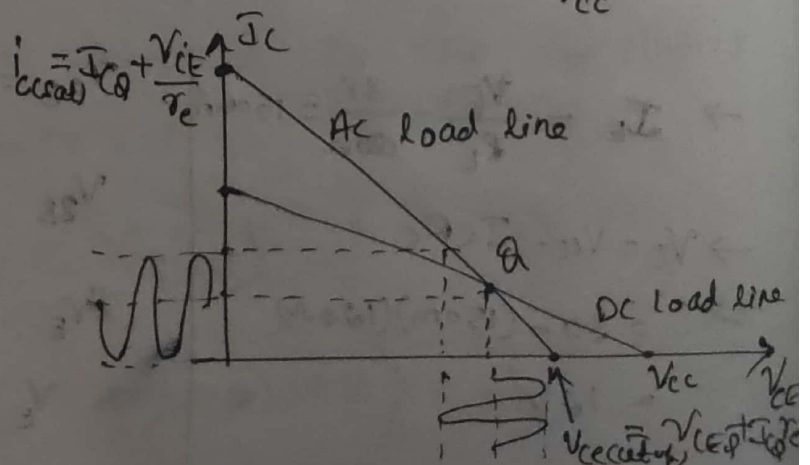
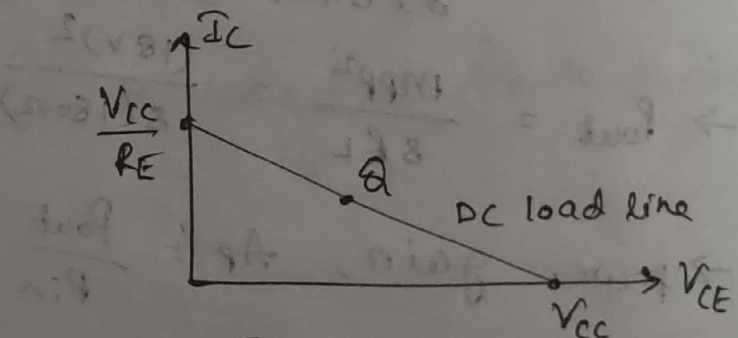
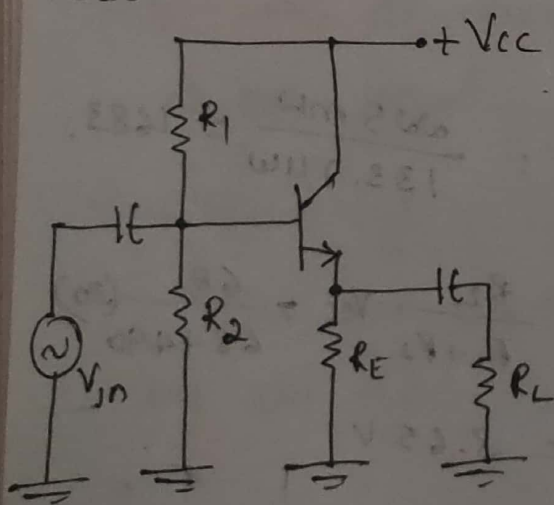
$$\therefore I_{DC} = 53.8mA + 150mA = 203.8mA$$

$$\rightarrow P_{DC} = V_{CC} I_{DC} = (30V)(203.8mA) = 6.11W$$

$$\rightarrow \eta = \frac{P_{out}}{P_{DC}} \times 100 = \frac{225mW}{6.11W} \times 100 = \underline{\underline{3.68\%}}$$

Emitter Follower Power Amplifier

The circuit is used to get the Q point at the center of load line. & is shown in fig below.



→ large values of R_2 will saturate transistor

$$\therefore \langle I_{C(sat)} = \frac{V_{CC}}{R_E} \rangle$$

→ small values of R_2 will drive transistor to cutoff.

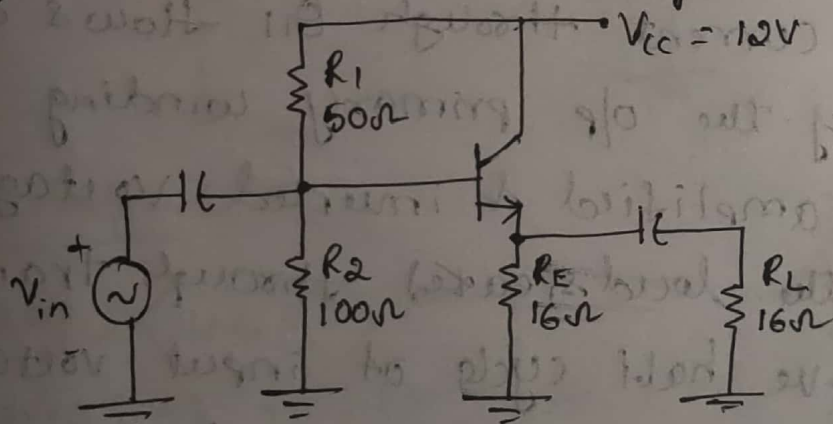
$$\therefore \langle V_{CE(cutoff)} = V_{CC} \rangle$$

→ for ac load line,

$$i_{C(sat)} = I_{CQ} + \frac{V_{CE}}{r_e}$$

$$\& V_{CE(cutoff)} = V_{CE} + I_{CQ} r_e$$

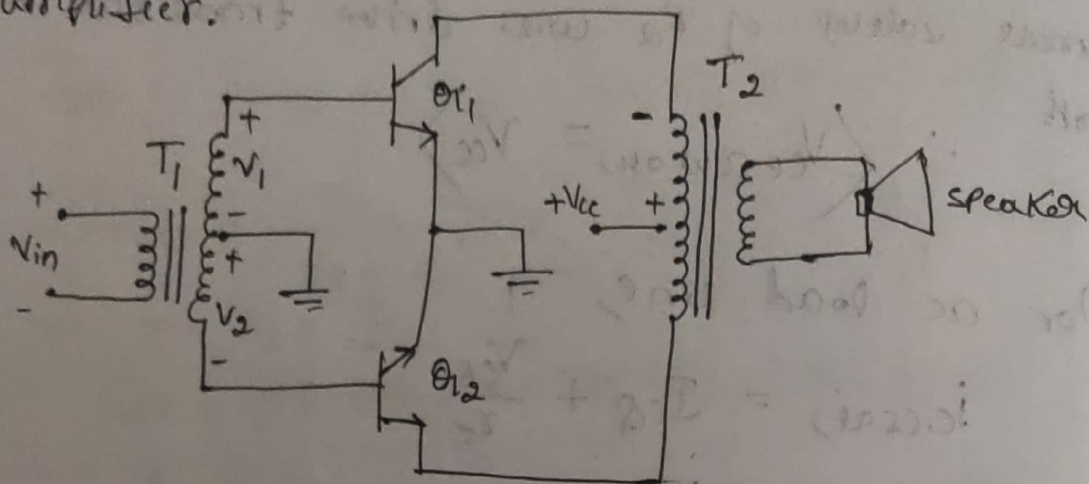
EX2:- For the circuit shown calculate I_{CQ} , V_{CEQ} , r_e & also determine ac saturation & cutoff points, mpp output voltage of the circuit.



$$\rightarrow I_{CQ} = \frac{V_{CC}}{R_E}$$

class B operation:

→ Figure below shows the class B push-pull Amplifier.



→ Working:

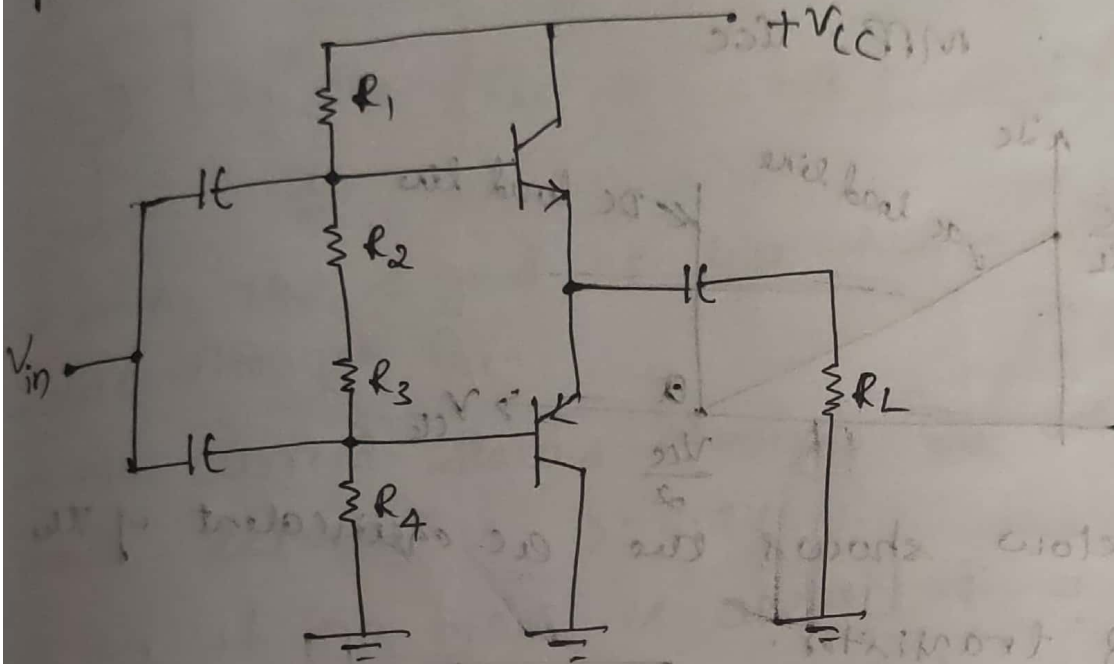
- * During +ve half cycle of input voltage, the secondary winding of T_1 has voltage V_1 & V_2
- * Hence Q_1 transistor conducts & Q_2 cuts off. & collector current through Q_1 flows during +ve half of the o/p primary winding which produces amplified & inverted voltage & coupled to the loudspeaker through transformer T_2
- * During -ve half cycle of input voltage, the Q_2 will turn on & Q_1 turns off. & Q_2 amplifies the signal & transformer-coupled to the loudspeaker.

* since each transistor amplifies one-half of the input cycle, the loudspeaker receives complete cycle of the amplified signal.

class B push-pull emitter follower;

→ In class B operation, the collector current flows only for 180° of the ac cycle.

→ Figure below shows the class B push-pull emitter follower which consists of both npn & pnp emitter follower in push-pull arrangement.



→ DC Analysis :

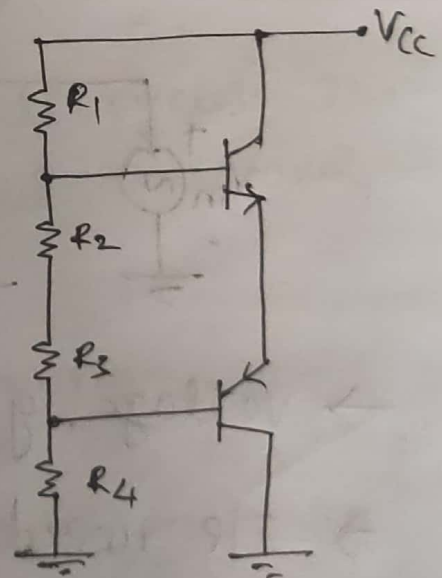
Ideally, $\langle I_{CQ} = 0 \rangle$

All biasing resistors are equal

$$R_1 = R_2 = R_3 = R_4$$

emitter diode is biased with same value of voltage.

$$\therefore \langle V_{CEQ} = \frac{V_{CC}}{2} \rangle$$



→ Since there is no dc resistance in the collector or emitter circuits from the diagram, the dc saturation current $[I_{C(sat)}]$ is infinite.

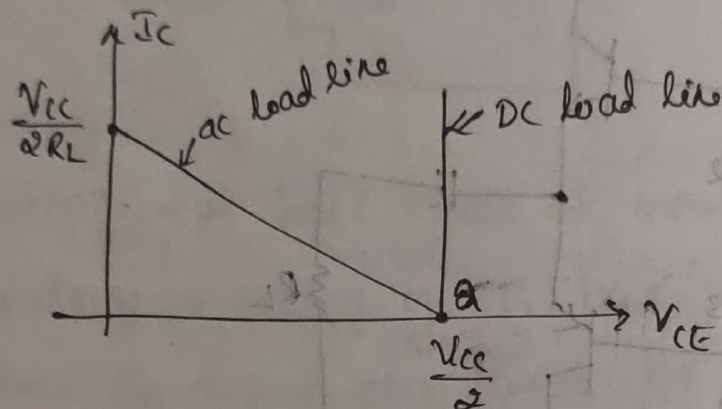
→ Hence DC load line is vertical as shown in fig. below.

→ AC load line:

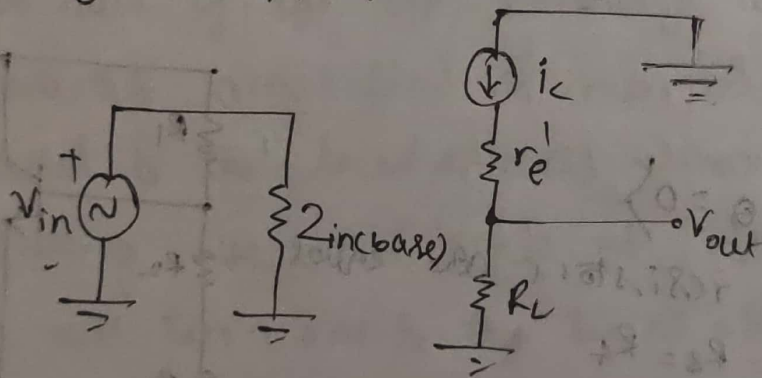
Fig. below shows the ac load line.

→ when either transistor is conducting, its operating point moves up along ac load line.

$$\therefore MPP = V_{CC}$$



→ Fig. below shows the ac equivalent of the conducting transistor.



→ Voltage gain, $A_V = 1$

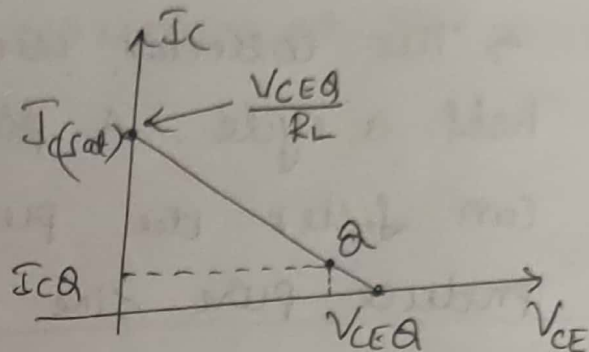
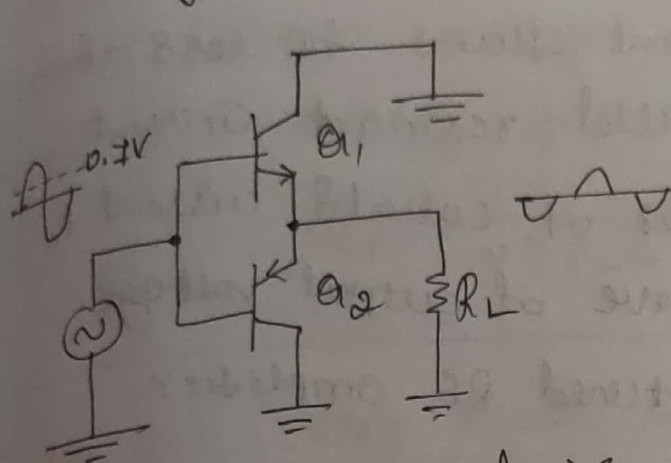
→ I/P impedance of the base is

$$Z_{in(base)} = \beta R_L$$

→

Crossover distortion in class B push-pull emitter follower.

→ fig. below shows the AC equivalent ckt.



→ when the signal is less than $0.7V$, no current flows through Q_1 .

→ No current flows through Q_2 until the ac input voltage is $> -0.7V$.

→ i.e., if no bias is applied to the emitter diodes, the output is clipped.

→ Because of clipping b/n half cycles, the output is distorted. It is called crossover distortion.

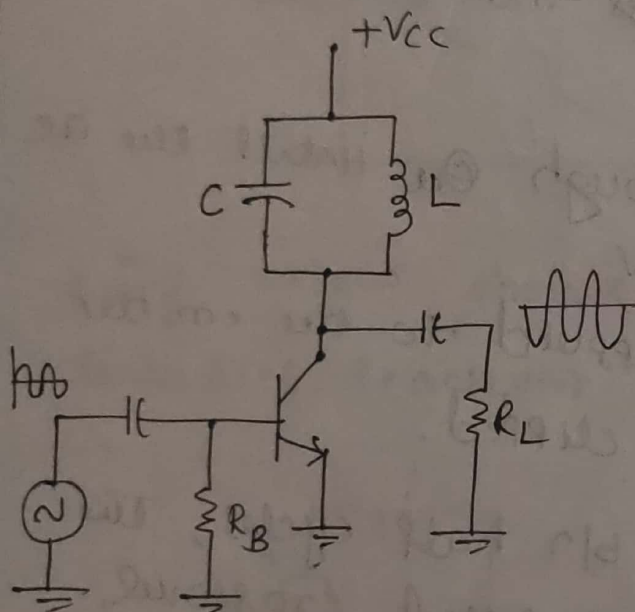
→ To eliminate crossover distortion, apply forward bias to each emitter diode i.e., locating Q point slightly above cutoff as shown in fig.

Class C operation / Tuned Amplifier

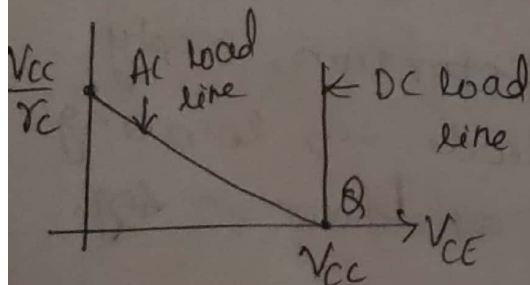
→ In class C operation, resonant circuit is used for load.

→ The collector current flows for less than half a cycle. A parallel resonant circuit can filter the pulses of collected current & produce pure sine wave of output voltage.

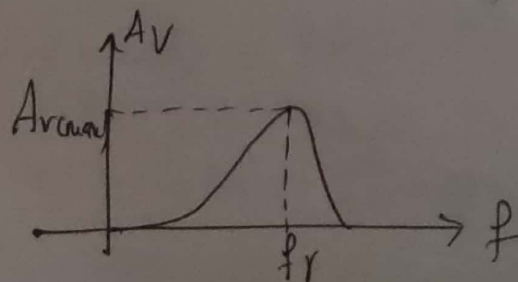
Fig. below shows tuned RF amplifier.



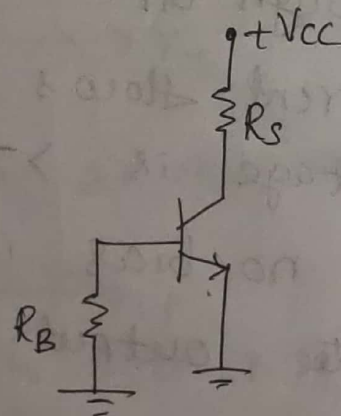
Two load lines



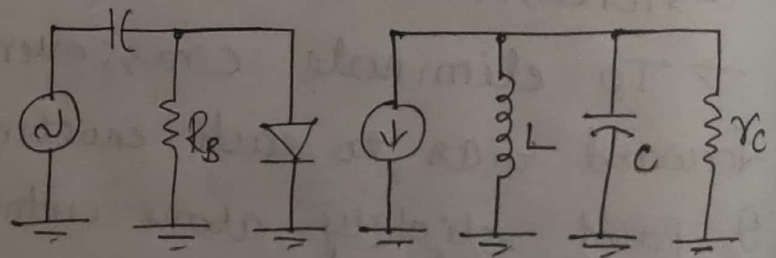
Voltage Gain



DC equivalent ckt



AC equivalent ckt



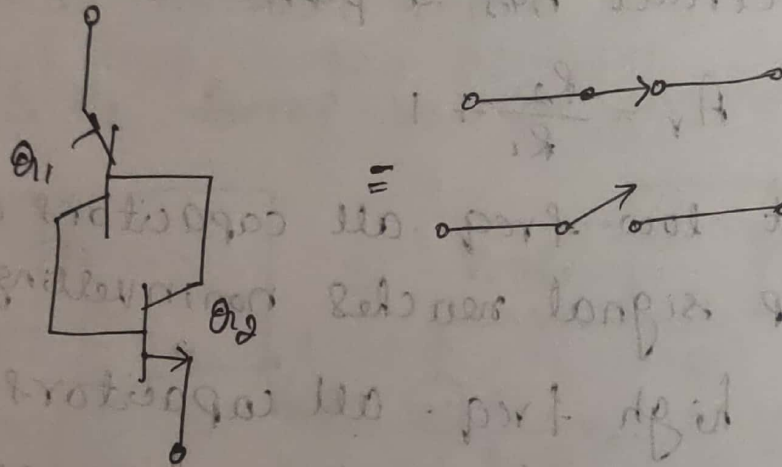
- The ac input voltage drives the base & an amplified output voltage appears at the collector.
- The amplified & inverted signal is capacitively coupled to the load resistance.
- The output voltage is maximum at the resonant frequency given by:

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

Thyristors

1) Four-layer Diode / Latching Circuit :-

→ Four-layer diode is shown in figure below. The upper transistor Q_1 is a PNP device & the lower transistor Q_2 is an NPN device. The collector of Q_1 drives the base of Q_2 . Similarly the collector of Q_2 drives the base of Q_1 .



→ Operation :

a) Positive feedback: The four-layer diode uses the feedback. i.e., any change in the base current of Q_2 is amplified & fed back through Q_1 . This the feedback continues changing the base current of Q_2 until both transistors go into either saturation or cutoff.

i) if the base current of Q_2 increases, the collector current of Q_2 increases, which in turn increases base current of Q_1 & collector current of Q_1 .

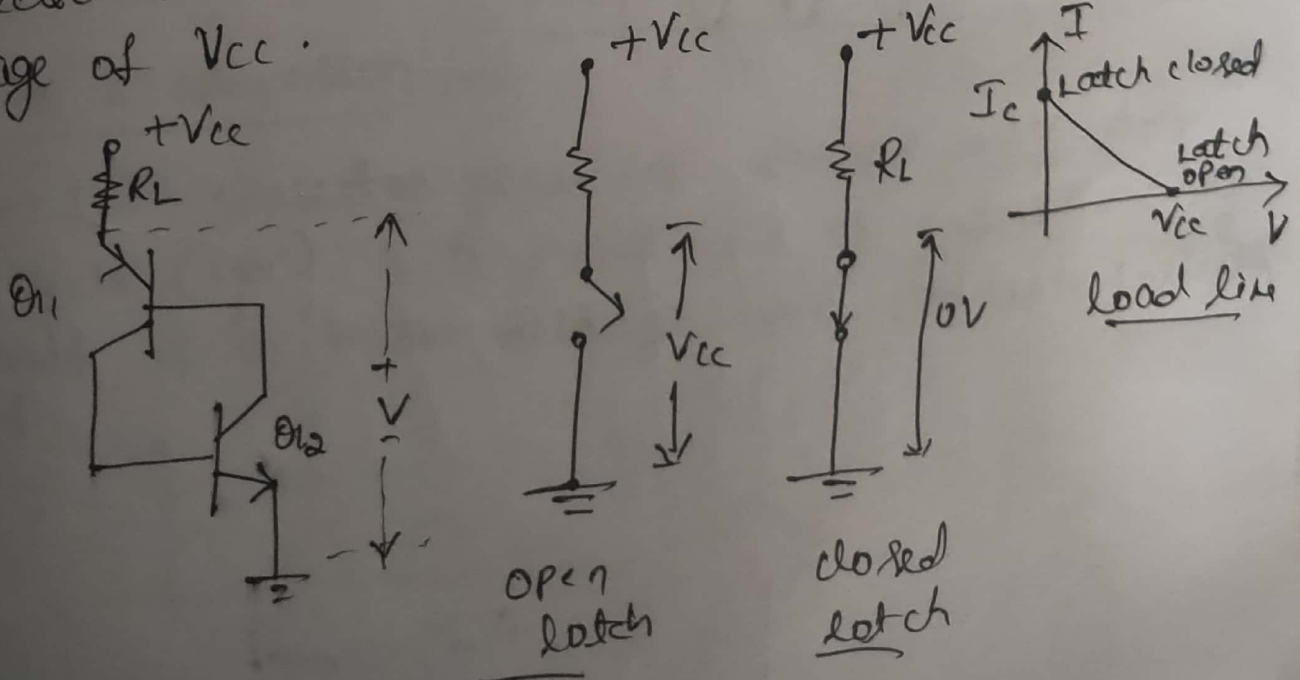
the collector current in Q_1 will further increase the base current of Q_2 . This amplify & feedback action continues until both transistors are driven into saturation, in this case the overall circuit acts like a closed switch.

i) if the base current of Q_2 decreases, the collector current of Q_2 decreases, the base current of Q_1 decreases, the collector current of Q_1 decreases & the base current of Q_2 decreases further. This action continues until both transistors are driven into cut-off. in this case the overall circuit acts like a open switch.

→ The circuit is stable in either closed or open. & it will remain in either state indefinitely until there is change in base current of Q_1 & Q_2 .

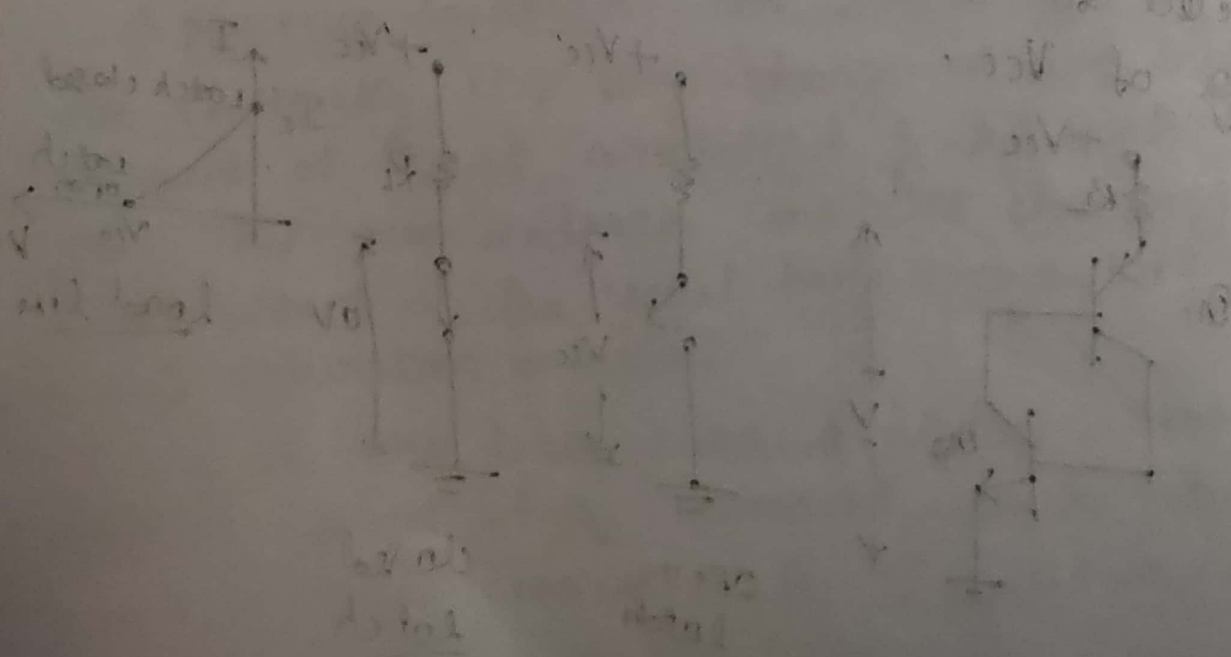
Hence the circuit is called latch.

i) closed latch : figure below shows a latch connected to a load resistor with a supply voltage of V_{CC} .



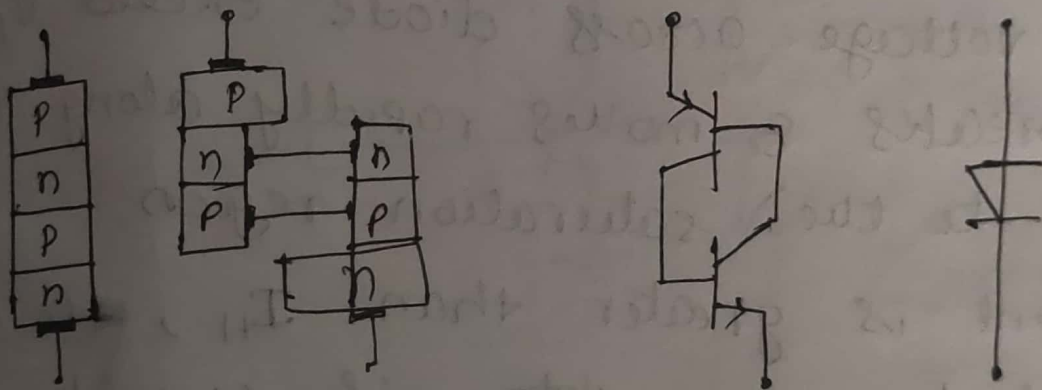
* Assume initially latch is open, no current through the load resistor, the voltage across the latch equals supply voltage. so the operating point is at the lower end of the load line.

* The only way to close the latch is by breakover. i.e., large enough supply voltage V_{ce} is applied to breakdown the Q_1 collector diode. since the collector current of Q_1 increases the base current of Q_2 , the feedback will start which drives transistors to saturation. the latch is closed. & operating point is at the upper end of the load line.



The Schottky Diode / four layer diode :- **

→ Fig. below shows the Schottky diode. it is also called four layer diode or pnpn diode or silicon unilateral switch.



→ It consists of two transistors

1) pnp transistor

2) npn transistor.

→ Fig. shows the schematic symbol of a four layer diode & is similar to latching circuit.

→ It can be closed by breakover condition,

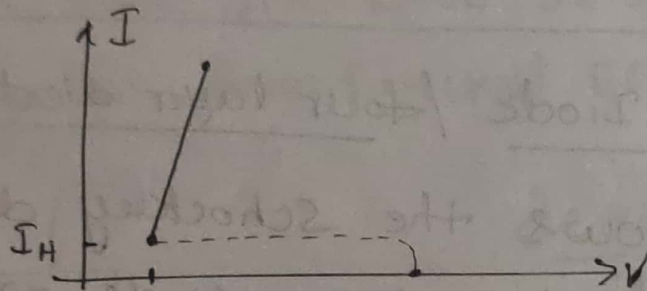
→ & to open it is by low-current drop-out i.e., reducing the current to less than the

holding current at which transistors switch from saturation to cutoff.

Breakover characteristic of four layer diode

→ fig below shows the graph of $I-V$ of four layer diode. The device has two operating regions: cutoff & saturation.

→ The dashed line is the transition path b/w cutoff & saturation.



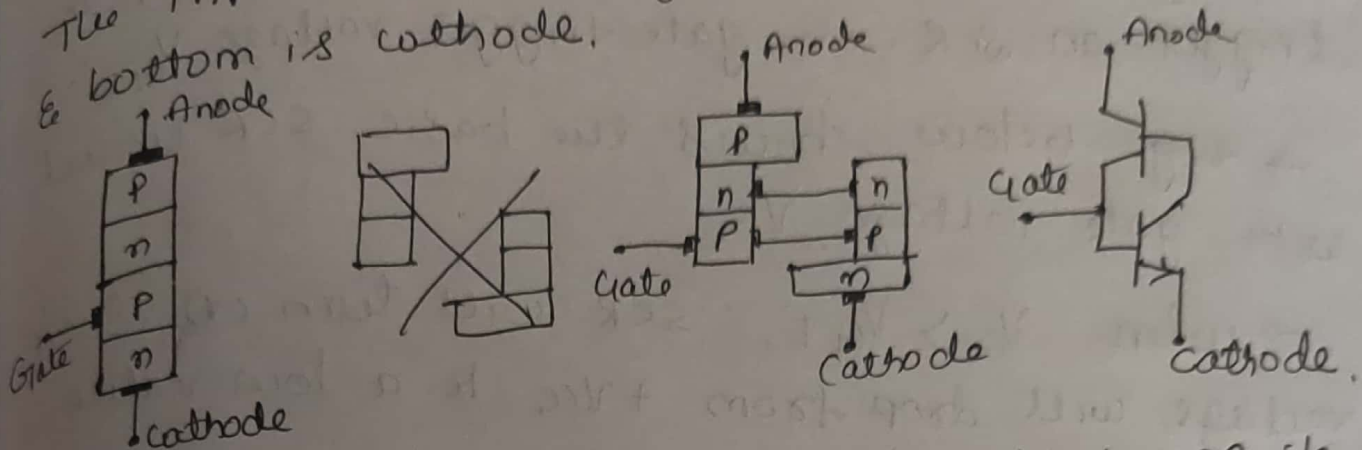
→ when the device is at cutoff, it has zero current.

→ If the voltage across diode exceed V_B , the device breaks & moves rapidly along the dashed line to the saturation region.

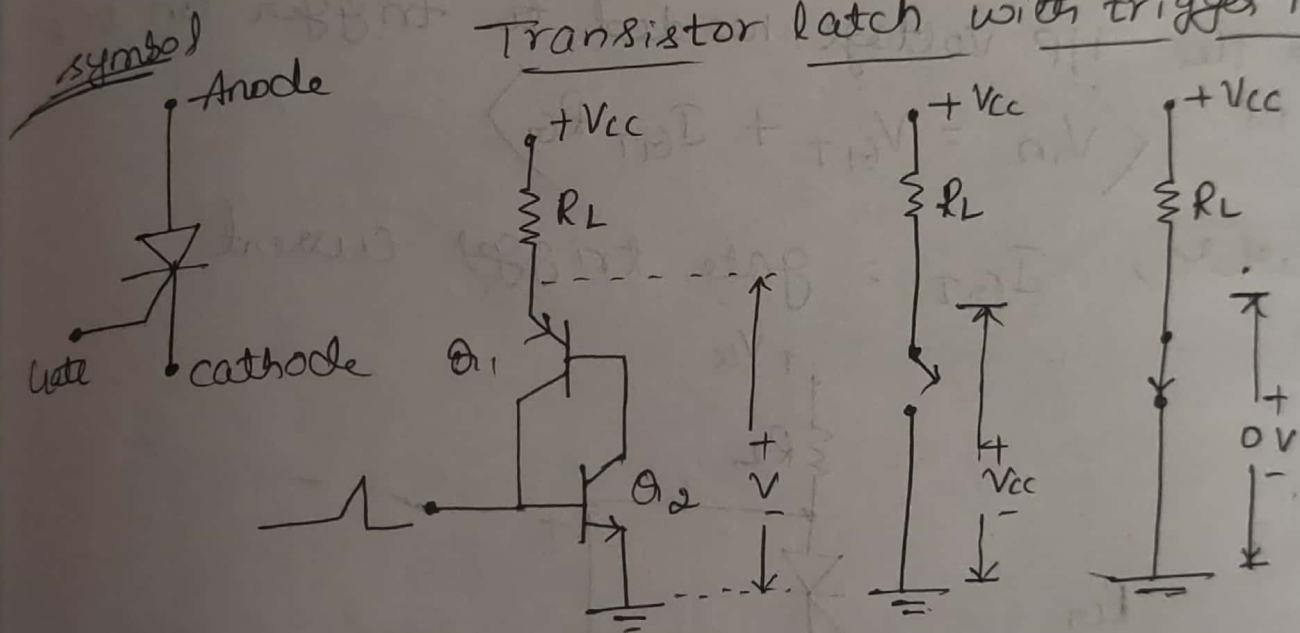
→ if current is greater than I_H , the diode remains latched in on state. if current, less than I_H , the device switches into cutoff.

Silicon controlled Rectifier

below shows the structure of SCR.
 → Fig. The input is called gate, the top is anode & bottom is cathode.



Transistor latch with trigger if

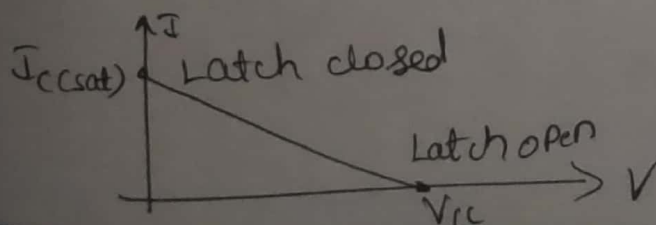


Working:

1) Triggering the latch

→ when the latch is open, the operating point is at the lower end of the dc load line.

→ To close the latch, a trigger pulse is coupled into the base of Q_2 as shown in fig. which increases base current of Q_2 .



➤ Gate triggering :

→ Gate of SCR is connected internally to the base of transistor, it takes at least 0.7V to trigger an SCR. i.e., gate trigger voltage V_{GT} .

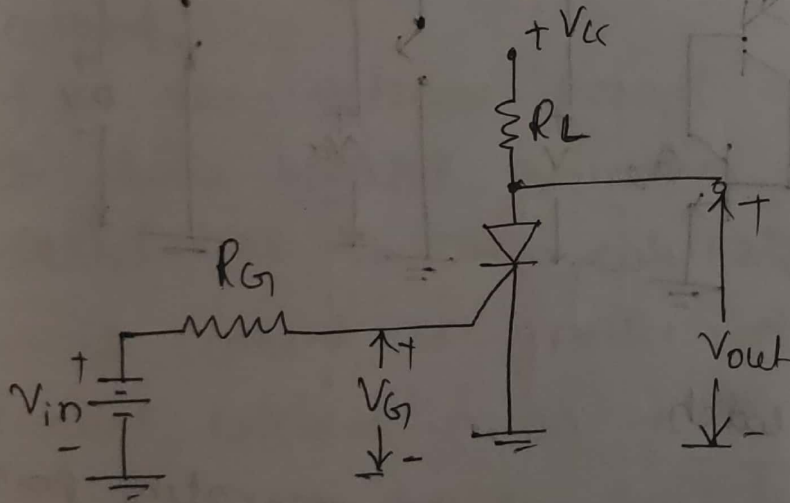
→ Fig. below shows the basic SCR circuit with gate voltage V_G .

→ when $V_G > V_{GT}$, SCR will turn on. E/o/p voltage will drop from $+V_{CC}$ to a low value.

→ The i/p voltage needed to trigger an SCR

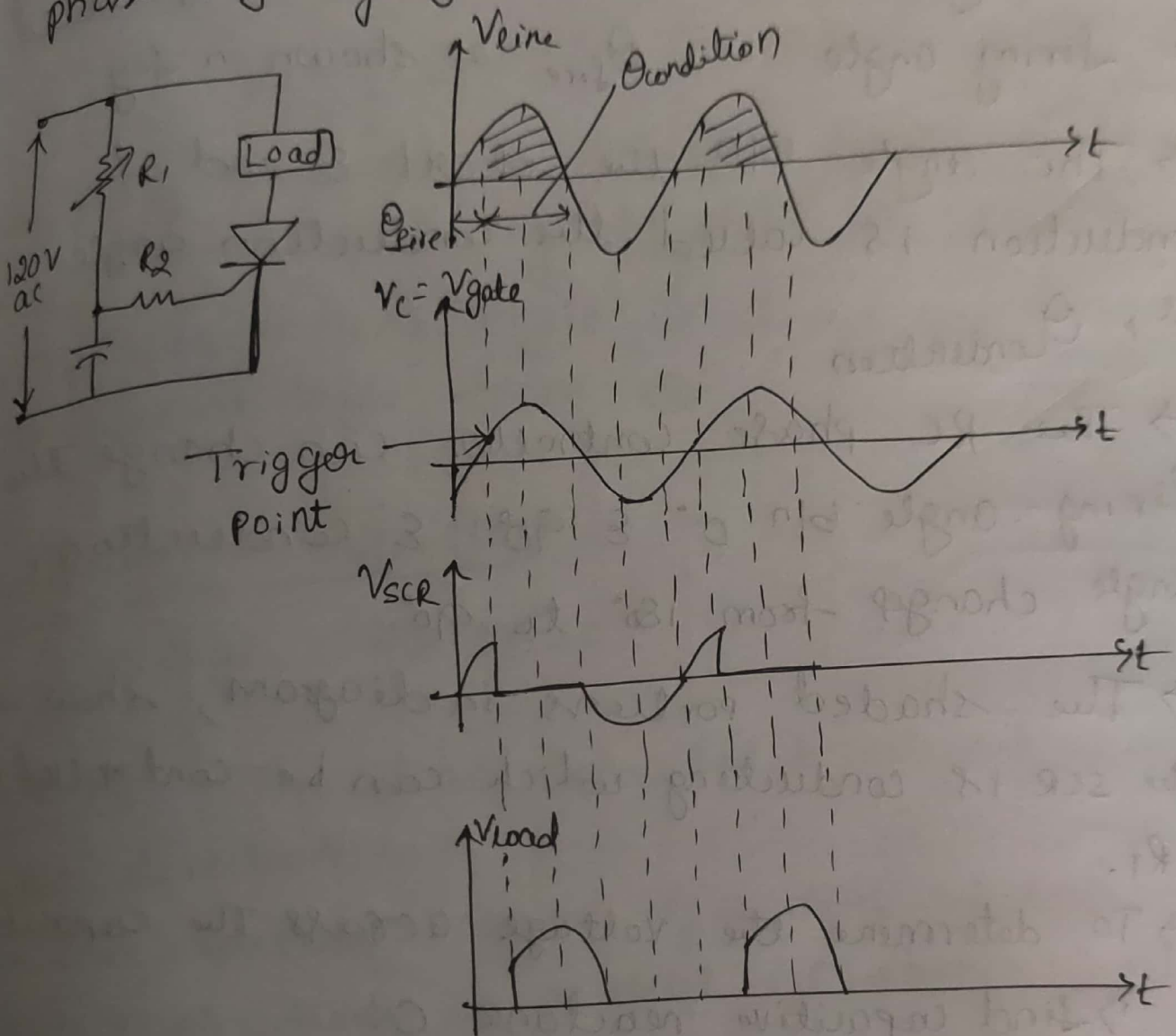
is
$$V_{in} = V_{GT} + I_{GT} R_G$$

where, I_{GT} = gate trigger current.



SCR phase control

→ Fig. below shows the SCR phase control which consists of RC circuit that controls the phase angle of gate signal.



→ when R_1 is zero, the gate voltage is in phase with the line voltage (V_{line}) & SCR acts like half-wave rectifier.

→ when R_1 increases, the ac voltage lags the line voltage by an angle ϕ 0° & 90° . as shown in fig.

i.e., before trigger point SCR is off & the load current is zero.

→ At the trigger point, the capacitor voltage is large enough to trigger the SCR. & V_{line} appears across load.

→ The angle at which the SCR fires is called the firing angle i.e., θ_{fire} as shown in fig.

→ The angle b/w the start & end of conduction is called the conduction angle i.e., $\theta_{conduction}$

→ The RC phase controller can change the firing angle b/w 0° & 90° & conduction angle changes from 180° to 90° .

→ The shaded portions in diagram, show when the SCR is conducting which can be controlled by R_1 .

→ To determine the voltage across the capacitor,
1) find capacitive reactance X_C .

$$X_C = \frac{1}{2\pi fC}$$

→ To find impedance & phase angle of the RC phase-shift circuit

$$Z_T = \sqrt{R^2 + X_C^2} \quad \theta_2 = \angle -\arctan \frac{X_C}{R}$$

3) i/p voltage as reference point, the current through C is

$$I_C \angle \theta = \frac{V_{in} \angle 0^\circ}{Z_T \angle -\arctan \frac{X_C}{R}}$$

4) Now, the voltage value & phase of the capacitor is

$$V_C = (I_C \angle \theta) (X_C \angle -90^\circ)$$

→ The amount of delayed phase shift will be approximate firing angle of the circuit.

→ The conduction angle is found by subtracting the firing angle from 180° .

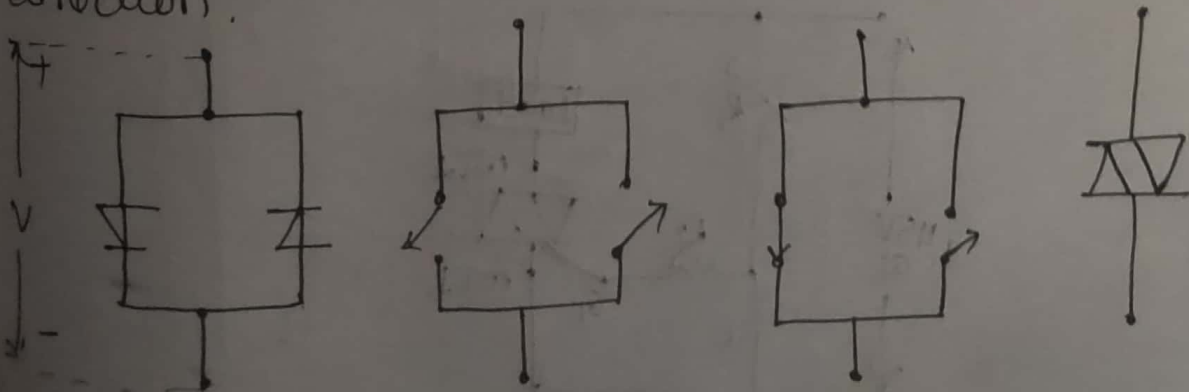
Bidirectional Thyristors:-

1) Diac 2) Triac.

1) Diac:- The diac can latch current in either direction.

→ The equivalent circuit of a diac is two four-layer diodes in parallel is shown in fig. below.

→ The diac is nonconducting until the voltage across it exceeds the break-over voltage in either direction.

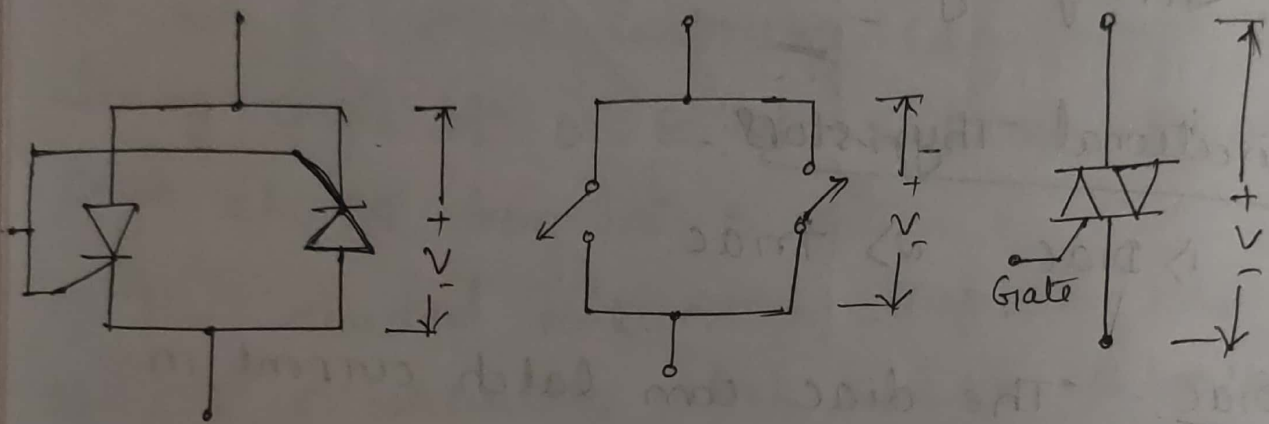


→ when left diode conducts when v exceeds the breakover voltage. & left latch closes.

→ when v has the opposite polarity, the right latch closes.

⇒ Triac

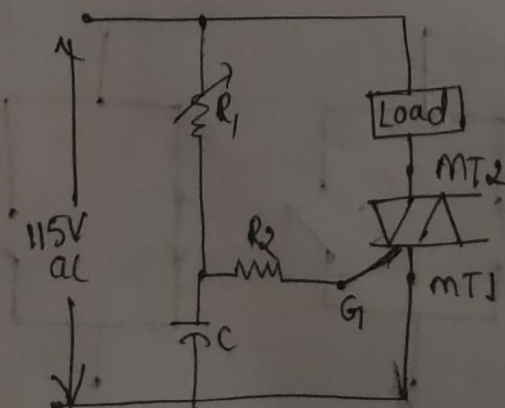
The triac acts like two SCR's in reverse parallel equivalent to the two latches as shown in fig. below.



→ Triac can control current in both directions.

→ If v has polarity, a +ve trigger will close the left latch. when v has opposite polarity, a negative trigger will close the right latch.

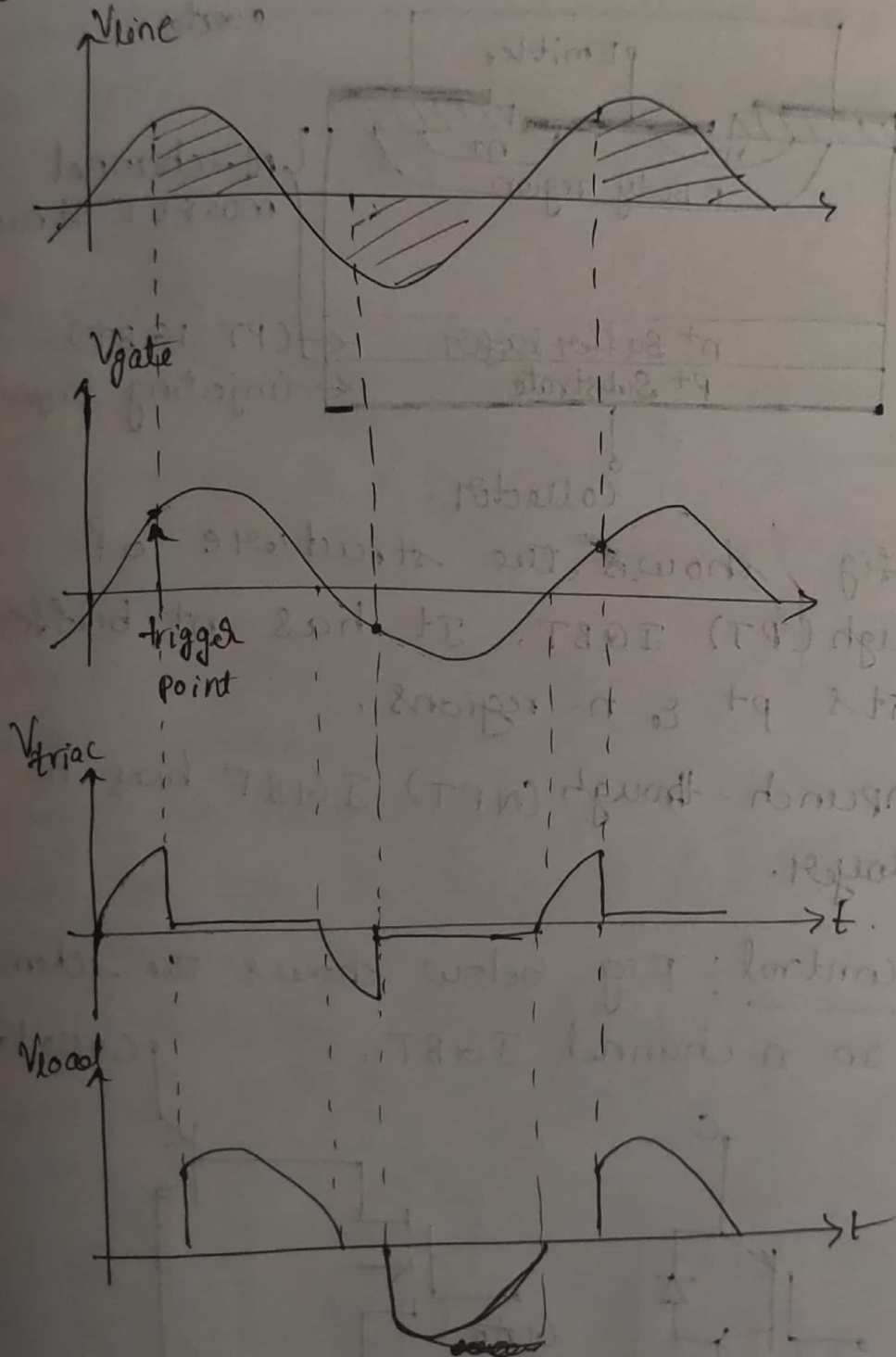
→ phase control: A Triac consists of RC circuit that varies the phase angle of the gate voltage to a triac



→ Figure below shows the line voltage & lagging gate voltage.

→ When the capacitor is large enough, to supply the trigger current, the triac conducts.

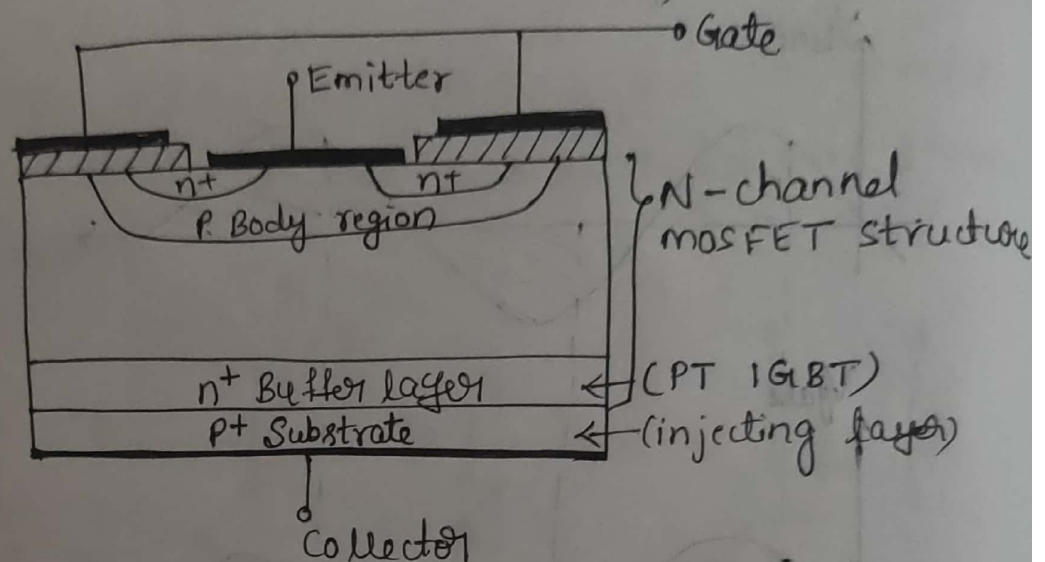
→ Once triac turns on, it continues to conduct until the line voltage returns to zero.



IGBT

Insulated Gate Bipolar Transistor

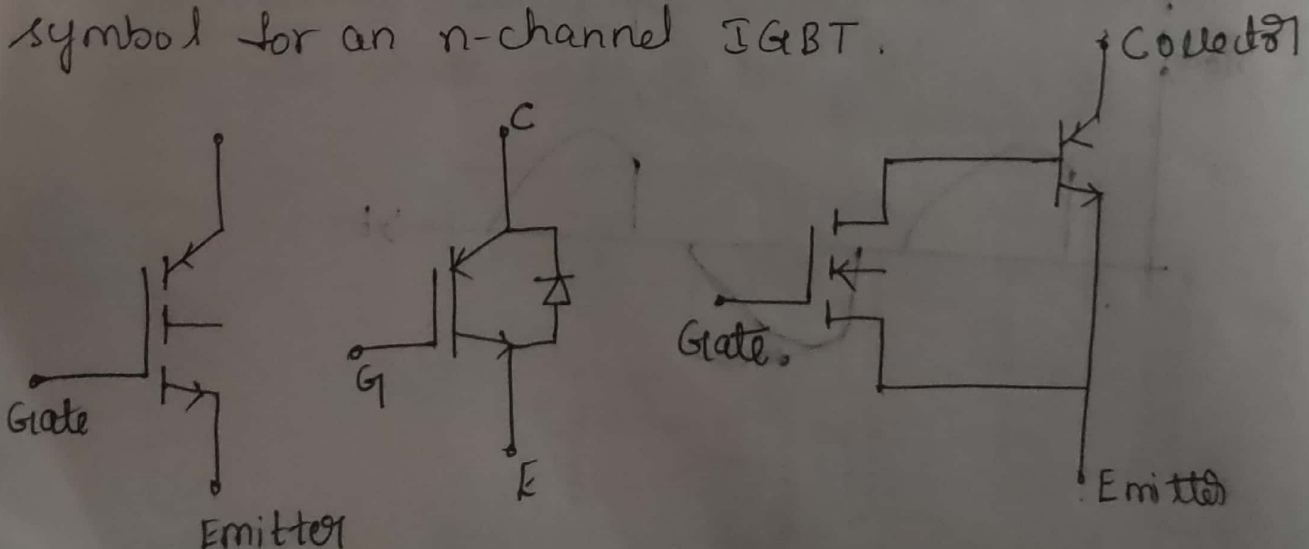
→ fig. below shows the basic structure of an n-channel IGBT, which is similar to an n-channel power MOSFET in a p-substrate.



→ Above fig. shows the structure of Punch-through (PT) IGBT. It has n+ buffer layer b/w its p+ & n- regions.

→ The nonpunch-through (NPT) IGBT has no n+ buffer layer.

→ IGBT control: Fig. below shows the schematic symbol for an n-channel IGBT.



- IGBT is essentially a power MOSFET on i/p side & a BJT on the output side.
- The i/p control is a voltage b/c the gate & emitter leads.
- The output is a current b/c the collector & emitter leads.
- When the input voltage, V_{GE} is large, collector current will begin to flow & this minimum gate vgl is called gate threshold voltage, $V_{GE(th)}$.

Other Thyristors:

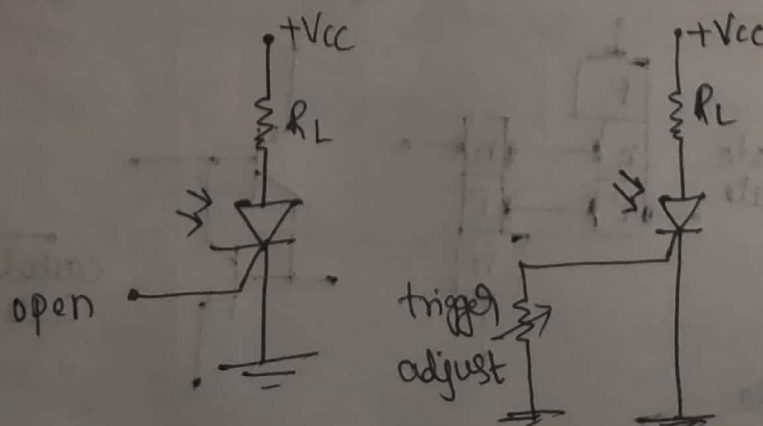
→ photo SCR: it is also called light-activated SCR.

→ The arrows in the figure represent the incoming light that passes through a window & hits the depletion layer.

→ When the light is strong enough, valence electrons ~~are~~ are dislodged from their orbits & become free electrons.

→ The flow of free electrons starts the +ve feedback & the photo-SCR closes.

→ After the light trigger has closed the SCR, it remains closed, even though the light disappears.

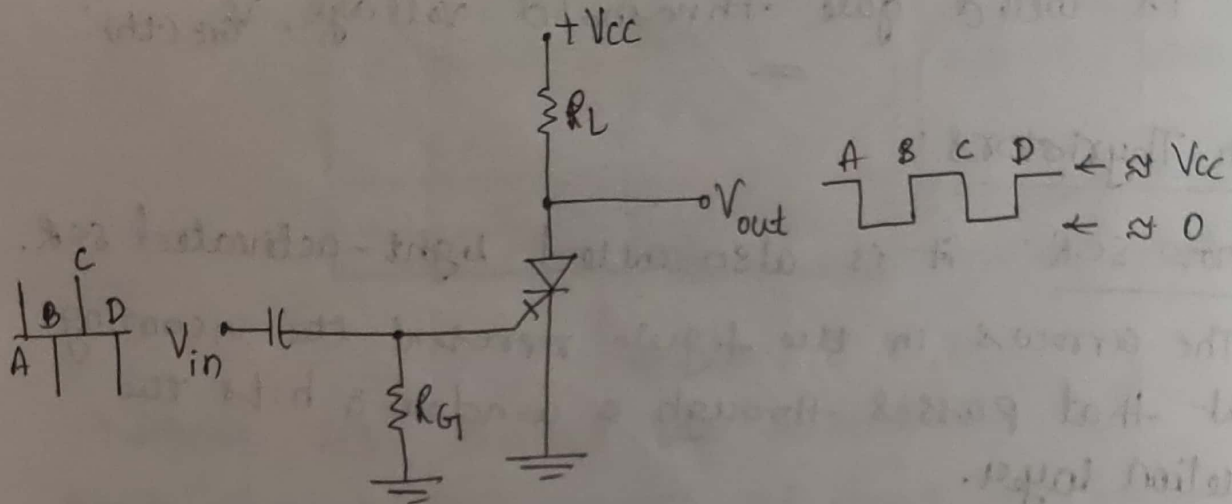


2) Gate Controlled Switch:

→ With gate controlled switch it is easy to open the switch with a reverse-biased trigger.

→ A gate controlled switch is closed by a positive trigger & opened by a negative trigger.

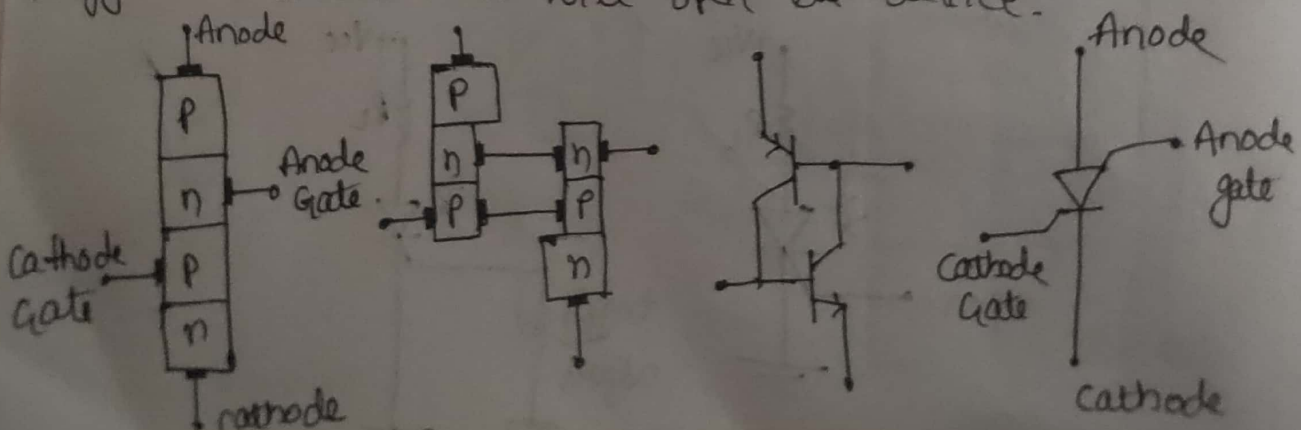
→ Fig below shows the gate controlled switch & its output is square waveform.



3) Silicon Controlled Switch:

→ Fig. below shows the doped regions of a silicon controlled switch. External lead is connected to each doped region.

→ A forward-bias trigger on either base will close the silicon controlled switch & a reverse-bias trigger on either base will open the device.



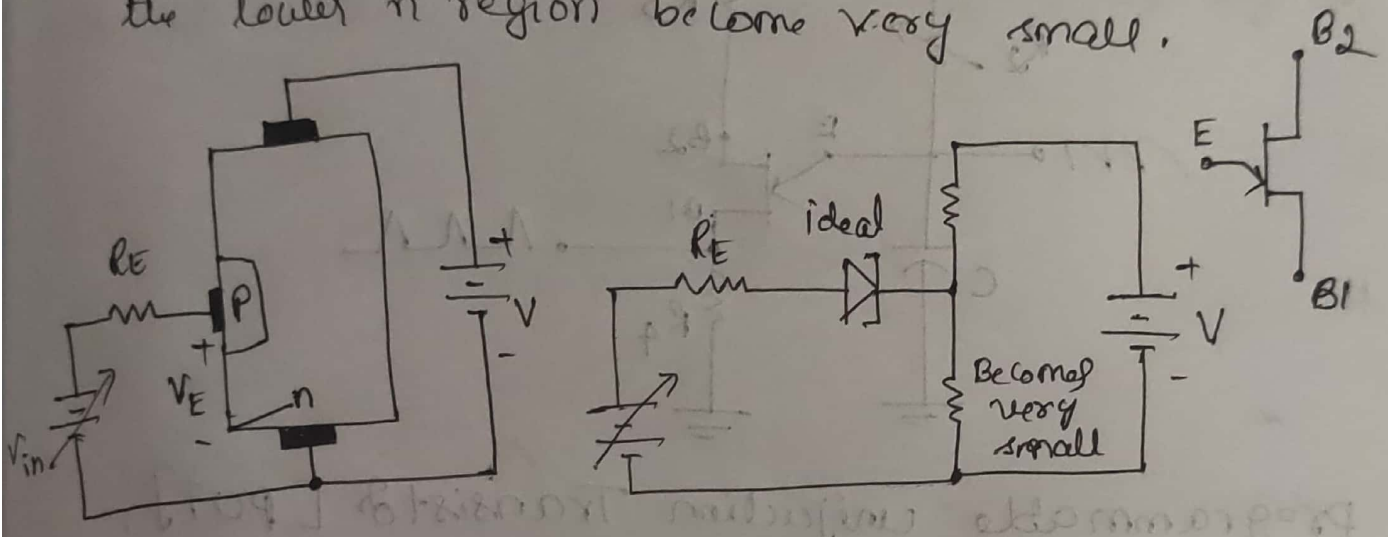
→ The lower gate is called cathode gate & upper gate is the anode gate.

4) Unijunction Transistor (UJT):-

→ Fig. shows UJT which has two doped regions.

→ When the i/p voltage is zero, the device is non conducting.

→ When we increase the input voltage above the standoff voltage, the resistance b/w p region & the lower n region become very small.



→ UJT relaxation oscillator:-

Fig. below shows the UJT relaxation oscillator i.e., a pulse generating circuit.

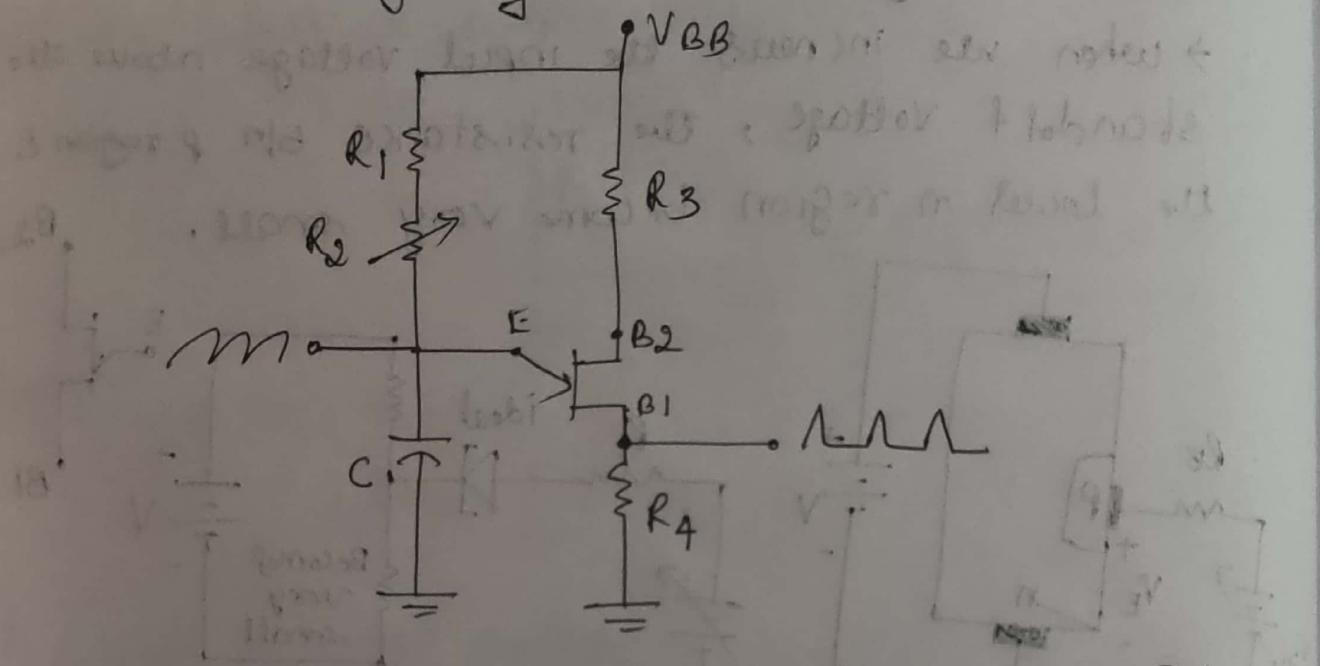
→ The capacitor charges towards V_{BB} .

→ When the capacitor voltage reaches a value equal to the standoff voltage, UJT turns on.

→ The capacitor discharges continuously due to internal low base resistance until low

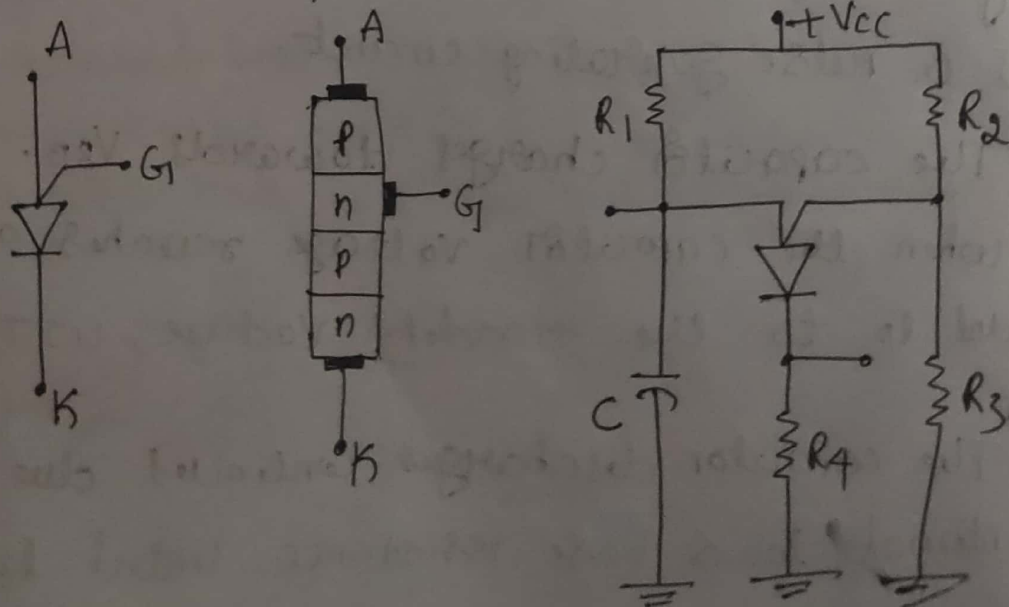
current dropout occurs. & UJT turns off.
 → Once again capacitor begins to charge towards V_{BB} .

→ The sharp pulse waveform developed across the external resistor at B_1 can be used as trigger source for controlling the conduction angle of SCR & triac circuit.



programmable unijunction Transistor [PUT]:-

→ Fig. below shows the four layer pnpn device used to produce pulses & waveforms.



→ The gate lead is connected to the n layer next to the anode & this pn junction is used to control the on & off states of the device.

→ When the anode voltage $\approx 0.7V$ higher than the gate voltage, the PUT turns on, remain in on state until its anode current falls below the holding current i.e., valley current I_v . & device will turn off.

→ The PUT is considered to be programmable because the gate voltage can be determined by an external voltage divider.